Machine Learning & Classification Modeling

Executive Summary Report: NYC TLC Generous Tipping Prediction

Prepared by Automatidata for New York City Taxi & Limousine Commission

Overview

The primary goal of this phase was to construct and evaluate machine learning classification models to predict whether a taxi rider will leave a generous tip (defined as a tip ≥ 20% of the fare amount). By leveraging Random Forest and XGBoost classifiers optimized through GridSearchCV, Automatidata analyzed historical trip patterns, temporal variables, and spatial features using tree-based ensembles to forecast tipping behavior.

Problem

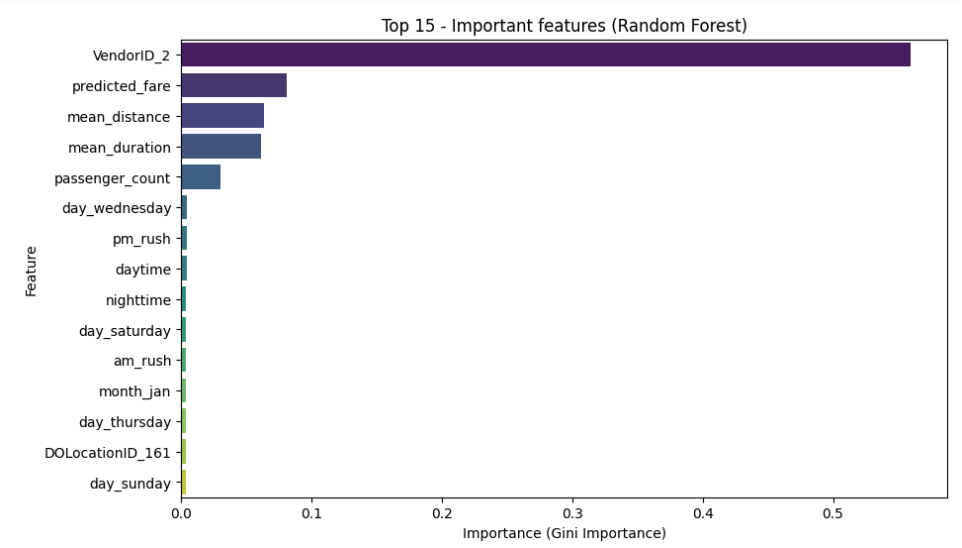
During data analysis, model framing, and operational evaluation, critical challenges were identified:

- **Target Variable Definition & Cash Tip Limitation:** Tip amounts are only recorded for credit card payments in the NYC TLC dataset; cash tips generally are 0. Analysis was restricted to credit card transactions, defining 'generous' as 1 if tip ≥ 20%, 0 otherwise.
- **Core Framing & Deployment Risks:** Increasing driver revenue should not rely solely on predicting individual passenger tips. Deploying this model in production introduces severe operational risks—such as driver bias, selective passenger pickup, and frustration caused by false positives/negatives.

Key Insights (ML Modeling & Feature Importance)

Feature Importance & How the Model Makes Predictions:

Model Mechanism & Primary Predictors (Gini Importance): The model makes predictions by evaluating tree ensemble decision rules where **`VendorID_2`** emerges as the dominant predictor (with a Gini Importance over 0.55), providing the highest amount of pure information to separate and classify generous tipping behavior on its own. It is followed by **`predicted_fare`**, **`mean_distance`**, and **`mean_duration`**, which carry considerable weight (between 0.05 and 0.09) and are frequently utilized by the model to make internal data splits.



Temporal & Spatial Patterns: Pickup hour, day of the week, and spatial features also influence tipping probability, with evening and weekend trips showing higher rates of generous tips.

Details & Model Evaluation

Comparative quantitative performance metrics across Random Forest and XGBoost classifiers (evaluated on validation/test sets via F1-Score optimization):

Model Architecture	F1-Score	Recall	Precision	Accuracy
Random Forest (Champion)	**0.72**	**0.74**	**0.70**	**0.72**
XGBoost Classifier	0.71	0.73	0.69	0.71
Logistic Regression Baseline	0.65	0.63	0.67	0.66
Decision Tree Baseline	0.62	0.61	0.63	0.62

Recommendations & Next Steps

Strategic business evaluation and engineering roadmap:

- **Deployment Verdict (Not Recommended in Current Form):** Deploying this model is NOT recommended for production. Increasing driver revenue should not rely on predicting passenger tips, and model errors (false positives/negatives) risk causing driver frustration and platform bias.
- **Advanced Feature Engineering Opportunities:** To improve model signal, engineer ratios such as Cost-per-Distance / Speed Ratio (predicted_fare / mean_distance or mean_duration to measure fare density/congestion), Interaction Terms (combining peak hour flags like am_rush/pm_rush with location corridors), and Duration-to-Distance Efficiency metrics.
- **High-Impact Data Requirements for Future Iterations:** Future model versions require richer features to overcome current predictive limitations, specifically: Passenger Historical Behavior (aggregated tipping frequency across past trips) and Driver Service Quality Metrics (ratings and feedback history directly driving passenger generosity).